

بسم الله الرحمن الرحيم

Omdurman Islamic university

Faculty of pharmacy

First Year

GENERAL CHEMISTRY

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Course Contents:

The Matter and Energy, Atomic Structure, Theories & Electronic Configuration, Chemical Bonding & Bonding Theory, The periodic law and periodic classification of elements & significance of elements C.S.N.H.O, Radiochemistry (isotopes) & calculation of halve lives , Inorganic Radio Pharmaceuticals ,Organometallic chemistry & its importance in pharmacy, Inorganic compounds of significance in pharmacy, Study of important Inorganic Gases, oxygen, nitrogen, nitric oxide, carbon dioxide, helium and ammonia, Selected reactions in inorganic chemistry in relation to pharmacy.

References:

- Nivaldo Tro, Chemistry: Structure and Properties, Pearson, Ed: 2015.
- Essentials of Inorganic Chemistry: For Students of Pharmacy, Pharmaceutical Sciences and Medicinal Chemistry, Katja A. Strohfeldt 2015.
- Inorganic Chemistry Catherine E. Housecroft and Alan G. Sharpe, second edition 2005.
- General Chemistry, Darrell D. Ebbing, Steven D. Gammon 9th 2007.

Preface:

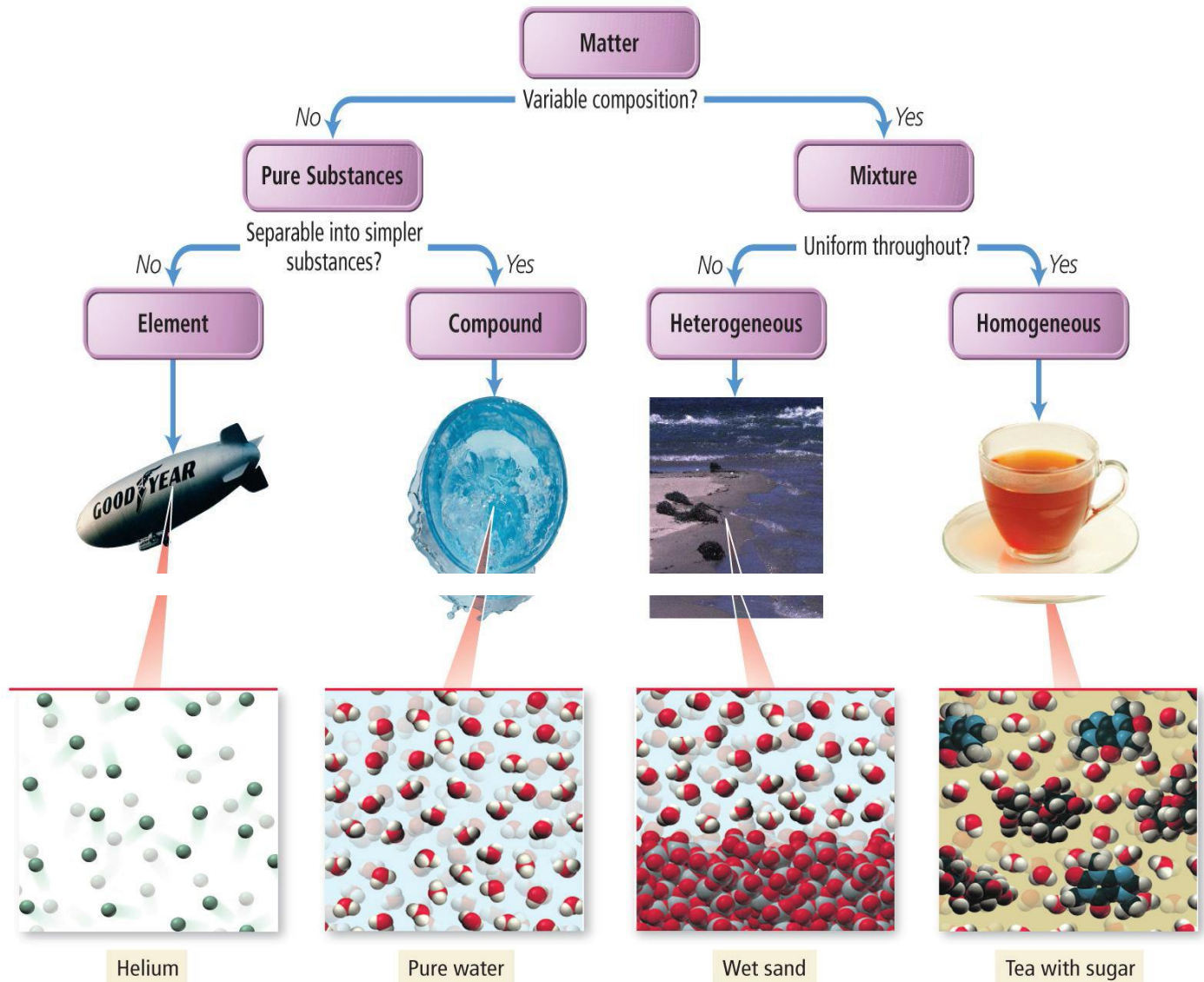
Chemistry is the study of composition, structure, properties and reaction of matter. Chemistry touches almost every aspect of our lives, our culture, and our environment. Its scope encompasses the air we breathe, the food we eat, the fluids we drink, our clothing, dwellings, transportation and fuel supplies, and our fellow creatures. Some call chemistry the central science. It rests on the foundation of mathematics and physics and in turn underlies the life sciences biology and medicine. To understand living systems fully, we must first understand the chemical reactions and chemical influences that operate within them.

The study of chemistry are divided into various branches are follows:

- **Organic chemistry:** The chemistry of compounds of carbon and hydrogen, Carbon is very versatile in its bonding and behavior and is a key element in many substances that are essential to life. All living matter contains carbon combined with hydrogen.
- **Inorganic chemistry** :The study of substances that do not contain carbon combined with hydrogen .
- **Analytical chemistry:** The branch of chemistry that is concerned with the detection or identification of substances present in a sample (qualitative analysis) or with the amount of each that is present (quantitative analysis) .
- **Physical chemistry** applies the mathematical theories and methods of physics to the properties of matter and to the study of chemical processes and the accompanying energy changes.
- **biochemistry** is the study of the chemistry of processes in living organisms

Matter: is any object that has a mass and occupies a space. Matter includes everything that is tangible, from our bodies and the stuff of our everyday lives to the grandest objects in the universe.

Types of matter : the following chart illustrates types of matter:



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Mixtures:

Two or more substances combined together, it has variable composition.

Classification of mixtures:

Mixtures can be classified into:

- **Homogeneous mixture:** having visibly indistinguishable Parts. (also known as a solution) is *a mixture that is uniform in its properties*

throughout given samples. When sodium chloride is dissolved in water, you obtain

a homogeneous mixture, or solution. Air is a gaseous solution, principally of two elementary substances, nitrogen and oxygen, which are physically mixed but not chemically combined

- **Heterogeneous mixture:** having visibly distinguishable parts. A heterogeneous mixture of salt and sugar is said to be composed of two different phases: one of the phases is salt; the other is sugar. Mixtures can be separated into pure substances by physical methods.

Pure matter: Has a definite composition and distinct properties.

Element: is a pure substance which cannot be chemically divided into a simpler substance. For example: iron (Fe), silver (Ag), , oxygen (O₂), hydrogen(H₂) etc. Elements can be classified as metals, non-metals or semi-metals.

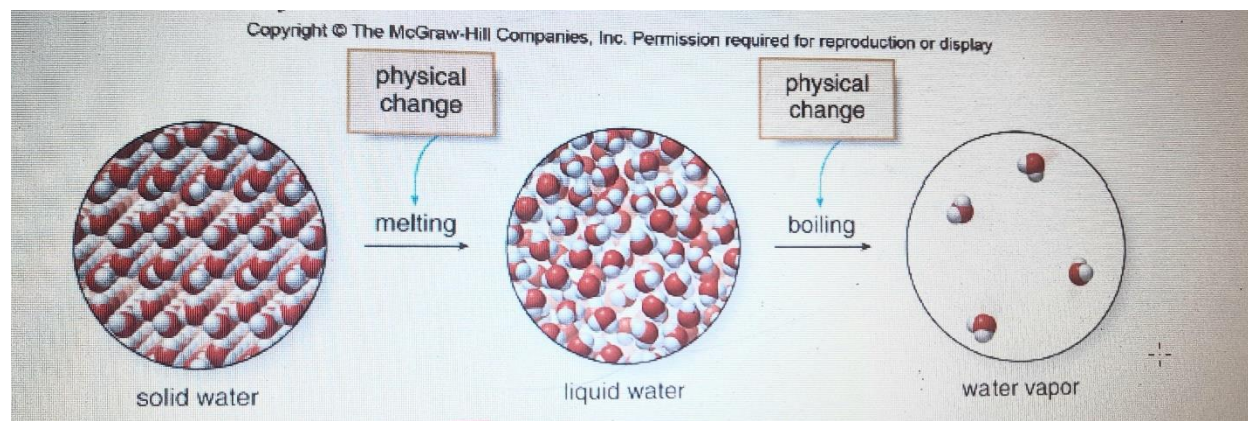
Compounds composed of two or more different elements chemically united in fixed proportions. A compound could be chemically divided into its constituent elements.

A **phase** is *one of several different homogeneous materials present in the portion of matter under study*

Matter changes: they are two types:

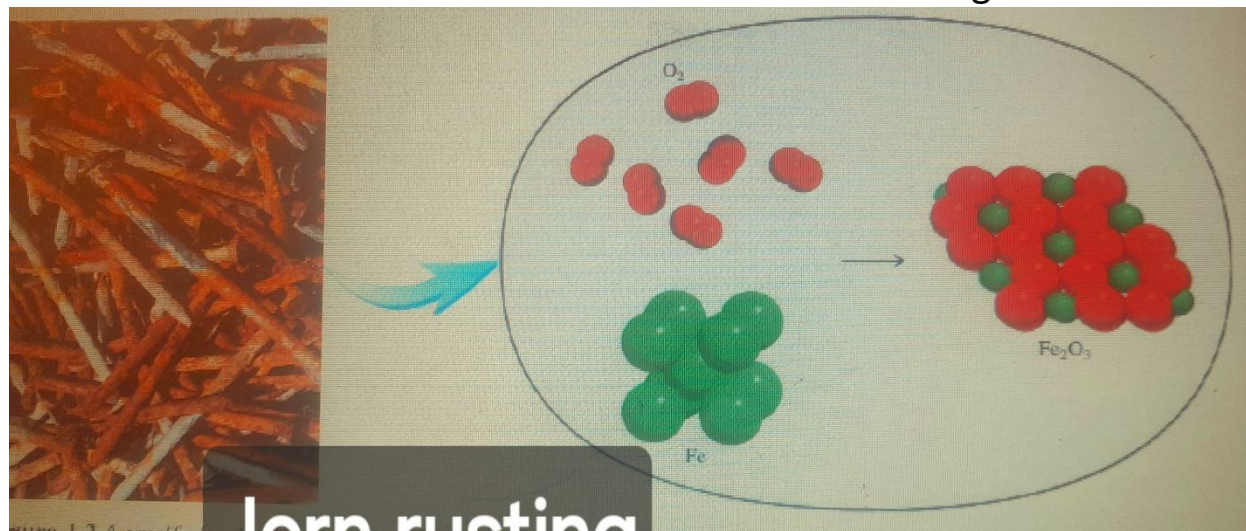
(1) Physical change:

Is a change in the form of a substance, not in its chemical composition, can be observed or measured without changing the composition of the material like boiling ,point,melting,point,solubility, color ,odor,state of matter



(2) Chemical change:

Chemical properties determine how a substance can be converted into another substance. Chemical change is the chemical reaction that converts one substance into another such as Iron rusting



Is one in which a given substance becomes a new substance or substances with different properties and different composition e.g. electrolysis of water to give hydrogen and oxygen.

State of Matter

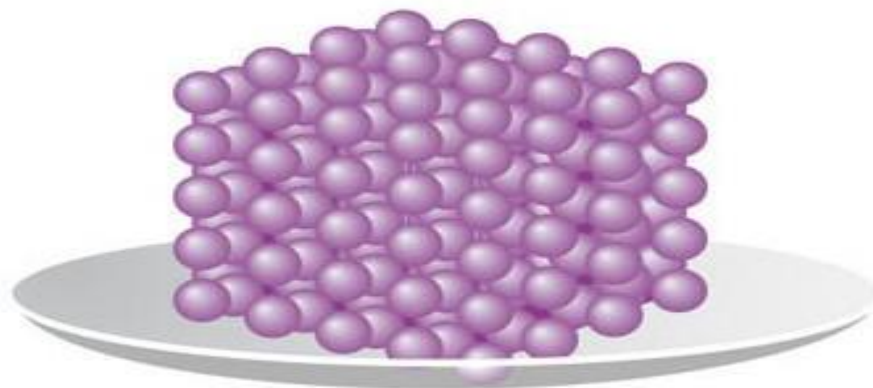
Matter exist in three physical states include

- i- Solid state
- ii- Liquid state
- iii- Gaseous state
- iv- Plasma.

Commonly, a given kind of matter exists in different physical forms under different conditions. Water for example, exists as ice (solid water), as liquid water, and as steam (gaseous water).

Solid state : *the form of matter characterized by rigidity* , Solid substance has definite shape and definite volume. Particle in solid substance are

closely packed together as shown in the diagram below. Examples of solid substances are Ice, metal, Wood, etc

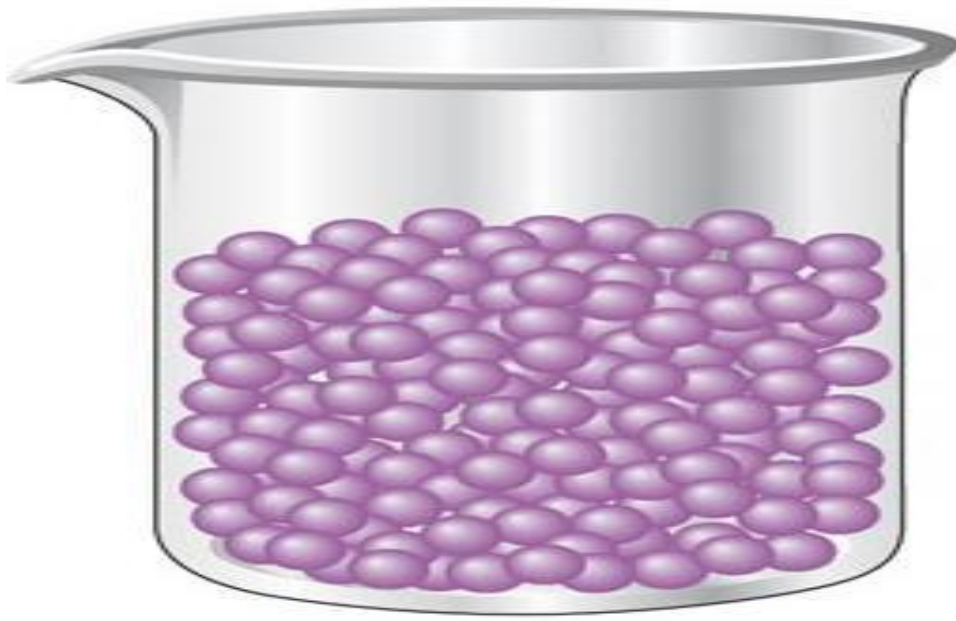


Solid

Properties of Solid Matter:

- i. Particles are closely packed together.
- ii. Has definite shape and volume.
- iii. Has strongest inter-particle force.
- iv. Particles are not free to move

Liquid State :*the form of matter that is a relatively incompressible fluid*; Liquid substance has fixed volume but variable in shapes. Particles in liquid are slightly farther apart. Examples: water, milk etc

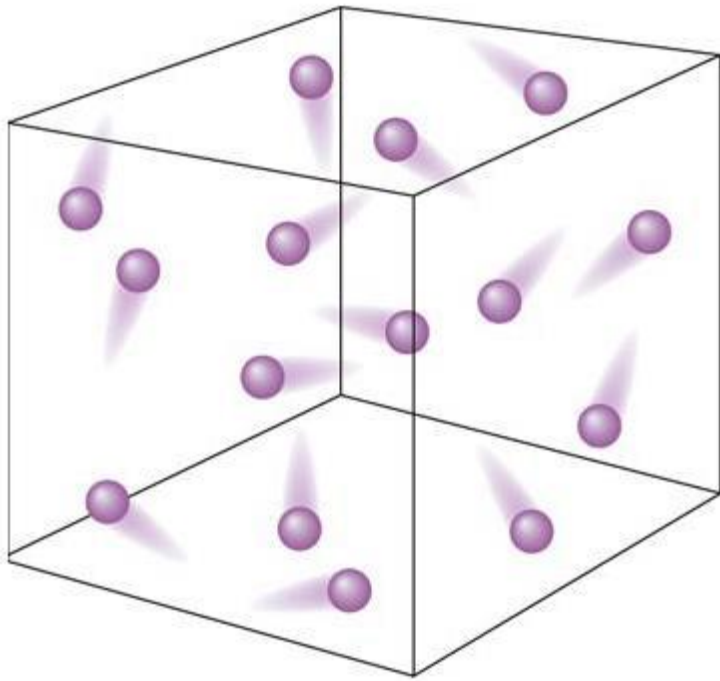


Liquid

Properties of Liquid Matter

- i. Particles are slight farther apart
- ii. Have definite volume
- iii. Have not definite shape
- iv. Has medium density due to medium volume
- v. Has medium motion
- vi. Inter-particle force is weak.

Gaseous State *the form of matter that is an easily compressible fluid* Gas has not definite shape or size. Particles are moving so fast and are so far apart that they do not interact with each other at all. For example oxygen, carbon dioxide etc.



Gas

Properties of Gas Matter:

- i. Has no definite shape
- ii. Has no definite volume.
- iii. Has largest inter particle distance.
- iv. Has low density due to largest volume.
- v. Has weakest inter particle force.
- vi. Particle move freely at high speed

Plasma State:

Plasma is an ionized gas consisting of a mixture of neutral molecules, atoms, positive ions, negative ions, free electrons, and other particles. These particles are formed as a result of sufficiently high temperatures that break bonds and dissociate molecules into ions.

Properties of the Plasma State:

- It has a very high temperature (10^3 - 10^6 K)

- It has high electrical conductivity due to the intense motion of charged particles.
- There is no clear surface boundary between plasma and gas, unlike the boundaries that exist between gas–liquid and gas–solid.
- It is called a conducting gas because it possesses the properties of a complete gas, represented in its state.
- It exists naturally in the outer regions of the Earth's atmosphere and in the atmospheres of stars, and it forms the beautiful aurora around the disk of the Sun and other stars.
- Its temperature is expressed in electron-volts ($1 \text{ eV} = 1100 \text{ K}$), and this temperature corresponds to the concept of the kinetic thermal energy of the particles present in the plasma, which makes it different from gases and not similar to them.